



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

CompTIA PenTest+ (PT0-003)

TGS Ref No: TGS-2024044563

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	3 July 2026	Initial release — CompTIA PenTest+ (PT0-003) 3-day lesson plan aligned to the 35 hands-on labs.	Dr. Alfred Ang
1.1	3 July 2026	Added Lab 12 (Wireshark/tcpdump) and Lab 21 (Netcat/Socat shells) — schedule now covers 37 labs across the three days.	Dr. Alfred Ang

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Course Information

Course Title	CompTIA PenTest+ (PT0-003)
WSQ Course Reference	TGS-2024044563
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	3 days · 8 training hours per day (24 hours)
Daily Timing	9:00 am – 6:00 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on penetration-testing labs on Kali Linux against authorised targets
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Plan and scope a penetration test — pre-engagement activities, SoW/NDA/RoE, threat modelling, MITRE ATT&CK mapping, CVSS risk scoring and professional reporting.
- LO2: Perform reconnaissance and enumeration — passive OSINT, DNS and subdomain discovery, internet-exposure intelligence, and active network, service and web enumeration.
- LO3: Discover and analyse vulnerabilities — network, web, container and secrets scanning, then validate findings and select public exploits safely.
- LO4: Execute attacks and exploits — network, authentication, application, wireless, cloud and social-engineering attacks against authorised targets.
- LO5: Conduct post-exploitation and lateral movement — persistence, credential-based lateral movement, pivoting, exfiltration testing, and cleanup with artifact preservation.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on penetration-testing tasks on Kali Linux, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 3 from 4:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Engagement Management & Reconnaissance

Time	Duration	Topic / Activity
9:00–9:30	30 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM)

9:30–10:30	60 min	Topic 1 — Engagement Management: pre-engagement, SoW/NDA/RoE, threat modelling, MITRE ATT&CK, CVSS and reporting (concepts + demo)
10:30–10:45	15 min	Tea break
10:45–13:00	135 min	Hands-on: Lab 1: Scoping a Pentest: SoW, NDA and Rules of Engagement; Lab 2: Threat Modeling with STRIDE and DREAD; Lab 3: Mapping a Pentest to MITRE ATT&CK; Lab 4: Risk Scoring with CVSS, CWE and EPSS; Lab 5: Writing a Penetration Test Report
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Topic 2 — Reconnaissance and Enumeration: passive OSINT, WHOIS/DNS, certificate transparency (concepts). Hands-on: Lab 6: Passive OSINT with theHarvester and Recon-ng; Lab 7: WHOIS, DNS Recon and Zone Walking; Lab 8: Subdomain Enumeration with Amass and Subfinder
15:30–15:45	15 min	Tea break
15:45–17:45	120 min	Hands-on: Lab 9: Shodan, Censys and Certificate Transparency; Lab 10: Network Scanning with Nmap; Lab 11: Service Enumeration with Nmap NSE; Lab 12: Traffic Capture and Analysis with Wireshark and tcpdump; Lab 13: Web Enumeration: Gobuster, Wfuzz, WhatWeb; Lab 14: Recon Scripting with Bash and Python
17:45–18:00	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Enumeration, Vulnerability Analysis & Exploitation

Time	Duration	Topic / Activity
9:00–9:15	15 min	Day 1 recap and mandatory digital attendance (AM)
9:15–10:30	75 min	Topic 3 — Vulnerability Discovery and Analysis: network, web, container and secrets scanning, finding validation (concepts + demo)
10:30–10:45	15 min	Tea break
10:45–13:00	135 min	Hands-on: Lab 15: Vulnerability Scanning with Greenbone /

		OpenVAS; Lab 16: Web Vulnerability Scanning with Nikto and ZAP; Lab 17: Container Image Scanning with Trivy and Grype; Lab 18: Secrets Scanning with TruffleHog; Lab 19: Validating Findings and Public Exploit Selection
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Topic 4 — Attacks and Exploits: Metasploit, credential attacks, privilege escalation (concepts). Hands-on: Lab 20: Exploitation with the Metasploit Framework; Lab 21: Bind and Reverse Shells with Netcat and Socat; Lab 22: LLMNR/NBT-NS Poisoning and SMB Relay with Responder
15:30–15:45	15 min	Tea break
15:45–17:45	120 min	Hands-on: Lab 23: Online Password Attacks with Hydra and Medusa; Lab 24: Offline Hash Cracking with hashcat and John; Lab 25: Linux Privilege Escalation with LinPEAS; Lab 26: Windows Privilege Escalation and Kerberoasting; Lab 27: SQL Injection with sqlmap
17:45–18:00	15 min	Day 2 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Attacks, Post-Exploitation & Assessment

Time	Duration	Topic / Activity
9:00–9:15	15 min	Day 2 recap and mandatory digital attendance (AM)
9:15–10:45	90 min	Hands-on: Lab 28: XSS, CSRF and SSRF with Burp Suite; Lab 29: File Inclusion, Directory Traversal and Web Shells; Lab 30: Cloud Attacks: S3 and IAM with Pacu and ScoutSuite; Lab 31: Wireless Attacks: WPA Cracking with Aircrack-ng; Lab 32: Social Engineering with GoPhish and SET
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Topic 5 — Post-exploitation and Lateral Movement (concepts). Hands-on: Lab 33: Establishing Persistence (cron, systemd, web shell); Lab 34: Lateral Movement with CrackMapExec and Impacket; Lab 35: Pivoting with sshuttle,

		Chisel and Proxychains
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Hands-on: Lab 36: Data Exfiltration over DNS, ICMP and HTTPS; Lab 37: Cleanup, Restoration and Artifact Preservation. Course recap and exam preparation
15:30–15:45	15 min	Tea break
15:45–16:00	15 min	Briefing for Assessment
16:00–17:00	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book
17:00–18:00	60 min	Practical Performance (PP) — hands-on penetration-testing tasks on Kali Linux, 1 hour, open book. PM digital attendance

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to exam skill areas)

Topic / Exam skill area	Weighting	Labs
Topic 01: Engagement Management	13%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5
Topic 02: Reconnaissance and Enumeration	21%	Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14
Topic 03: Vulnerability Discovery and Analysis	17%	Lab 15, Lab 16, Lab 17, Lab 18, Lab 19
Topic 04: Attacks and Exploits	35%	Lab 20, Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27, Lab 28, Lab 29, Lab 30, Lab 31, Lab 32
Topic 05: Post-exploitation and Lateral Movement	14%	Lab 33, Lab 34, Lab 35, Lab 36, Lab 37